# Jaeyeon Park

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## Education

**Seoul National University** — Seoul, Republic of Korea
*Ph.D. Candidate, Electrical and Computer Engineering* — *2023 – present*

- Lab: Power Conversion and Systems Lab (PCSL)
- Advisor: Prof. Shenghui Cui
- Research: Modular multilevel converters (MMC), medium-voltage variable frequency drives, power-hardware-in-the-loop (PHIL) testing, real-time simulation.

**Seoul National University** — Seoul, Republic of Korea
*M.S., Electrical and Computer Engineering; GPA: 4.03/4.3* — *2021 – 2023*

- Lab: Electrical Engineering & Power Electronics Lab (EEPEL)
- Advisor: Prof. Seung-Ki Sul
- Thesis: Real-Time Torque Control of PMSM under Flux Weakening Operation Based on Nonlinear Optimization.

**Seoul National University** — Seoul, Republic of Korea
*B.S., Electrical and Computer Engineering; GPA: 3.35/4.3* — *2015 – 2021*

- Thesis: Battery Electric Vehicle Motor Control System Design and Implementation.

## Research Experience

**Power Conversion & Systems Lab, Seoul National University** — Seoul, Republic of Korea
*Ph.D. Candidate; Advisor: Prof. Shenghui Cui* — *2023 – present*

- MMC-Based MV Variable Frequency Drive: Led development of control algorithms and control systems for a 5 MW Modular Multilevel Converter VFD for shipboard applications.
- PHIL Testing for Megawatt-Scale MMC: Developed a partial power-hardware-in-the-loop framework for in-situ dynamic and steady-state validation of individual submodules in a megawatt-scale MMC.

**Electrical Engineering & Power Electronics Lab, Seoul National University** — Seoul, Republic of Korea
*M.S. Researcher; Advisor: Prof. Seung-Ki Sul, IEEE Fellow* — *2021 – 2023*

- IPMSM Optimal Control: Developed online MTPA tracking and flux-weakening torque control algorithms for interior permanent magnet synchronous motors using sequential quadratic programming (SQP) and min-max optimization, achieving robust performance under parameter uncertainties without lookup tables.

**Signal Processing Systems Lab, Seoul National University** — Seoul, Republic of Korea
*Research Assistant; Advisor: Prof. Wonyong Sung, IEEE Fellow* — *2020 – 2021*

- Snoring Diagnosis Application: Developed custom ESP32 board and custom firmware for snoring diagnosis.

**Republic of Korea Army** — Seoul, Republic of Korea
*Research Engineer* — *2018 – 2020*

- Parallel Computing for Signal Processing: Developed parallel computing algorithms for massive signal processing.

## Teaching Experience

**Seoul National University** — Seoul, Republic of Korea
*Teaching Assistant, Electric Machines and Control; Instructor: Prof. Seung-Ki Sul* — *Fall 2021, Fall 2023*

- Responsibilities: Conducted lab sessions and tutorials; assisted students with coursework and problem sets.

*Lecturer, SNU Power Electronics Center Summer School* — *Summer 2022*

- Responsibilities: Delivered lectures on digital signal processor programming and embedded firmware development for power electronics applications.

## Publications

### Journal Articles

[1] D.-J. Kim, **J. Park**, Adrià Junyent-Ferré, and S. Cui, "Small-Signal Stability Analysis on Energy-Balancing Control of Modular Multilevel Converter under Instantaneous Optimal Zero-Sequence Voltage Injection," in *IEEE Transactions on Power Electronics*, accepted for publication.

[2] S.-Y. Lee, **J. Park**, D. Kim, S.-K. Sul, S. Cui, and J.-J. Jung, "Megawatt-Scale Partial PHIL Arm-Current Emulation for Submodule-Level Verification of MMC-Based Drives," in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, doi: 10.1109/JESTPE.2026.3719385.

[3] **J. Park**, D.-J. Kim, S. Lee, and S. Cui, "Improved Decoupled Control of Modular Multilevel Converter under Constraint of Nearest Level Modulation via Disturbance Observer Design," in *IEEE Transactions on Power Electronics*, doi: 10.1109/TPEL.2026.3684833.

[4] D.-J. Kim, **J. Park**, Y. Hong and S. Cui, "Fault-Tolerant Independent Pole Control Scheme of Zigzag MMC for Bipolar MVDC Distribution Systems," in *IEEE Transactions on Power Electronics*, vol. 41, no. 6, pp. 10324-10342, June 2026, doi: 10.1109/TPEL.2026.3655014.

[5] **J. Park**, D.-J. Kim, S.-Y. Lee, J.-J. Jung, S.-K. Sul and S. Cui, "Partial Power-Hardware-in-the-Loop Test for In-Situ Dynamic and Steady-State Validation of a Single Submodule in a Megawatt-Scale MMC," in *IEEE Transactions on Power Electronics*, vol. 41, no. 5, pp. 8409-8427, May 2026, doi: 10.1109/TPEL.2025.3642797.

[6] J. Maeng, J. Ham, G. Park, **J. Park**, H. Wu, J.-J. Jung, S. Cui, "A Supercapacitor Size Minimization and Energy Management Strategy for E-STATCOM Connected to Weak Grid," *International Journal of Electrical Power & Energy Systems*, vol. 172, Art. no. 111174, Nov. 2025, doi: 10.1016/j.ijepes.2025.111174.

[7] **J. Park** et al., "Equivalent Circuit Extraction of MI-HTS Coil Including High-Frequency Region for Electromagnetic Transient Simulation With Voltage-Source Converter," in *IEEE Transactions on Applied Superconductivity*, vol. 35, no. 7, pp. 1-11, Oct. 2025, doi: 10.1109/TASC.2025.3587992.

[8] Y.-W. Kim, H. Kim, **J. Park**, S.-K. Sul, "On-line Minimum Loss Control Strategy of IPMSM in Torque-Controlled Application," *Journal of Power Electronics*, vol. 24, p. 756, Apr. 2024, doi: 10.1007/s43236-024-00773-w.

[9] **J. Park**, J. Lee, J. Maeng, and S. Cui, "Sub-Switching-Period Current Limiting via PWM Carrier Shifting under Grid Faults," in *IEEE Transactions on Power Electronics* (under review).

### Conference Papers

[1] **J. Park** and S. Cui, "Multicore Simulation for Large-Scale Power Electronics Circuits in PLECS," *27th IEEE Workshop on Control and Modeling for Power Electronics (COMPEL)*, July 2026.

[2] S. Lee, **J. Park**, D. Kim and S. Cui, "Optimal Nearest-Level Modulation of MVDC-MMC with Decoupled Control for Lower Phase Current Total Harmonic Distortion," *2025 IEEE Energy Conversion Congress and Exposition (ECCE)*, Philadelphia, PA, USA, 2025, pp. 1-5, doi: 10.1109/ECCE58356.2025.11259672.

[3] J. Yun, **J. Park**, S. Moon, G. De Carne and S. Cui, "Resource-Efficient Real-Time Simulation of Multi-Active Bridge Converters Using Modular Programming," *2025 IEEE Kiel PowerTech*, Kiel, Germany, 2025, pp. 1-6, doi: 10.1109/PowerTech59965.2025.11180564.

[4] **J. Park**, J. Lee, D. Kim, S. Lee, S. Moon, J. Yoo, S. Ko, S. Cui, "MMC-Based High-Power Motor Emulator for Testing of Electrical Propulsion System," *2025 IEEE Electric Ship Technologies Symposium (ESTS)*, Alexandria, VA, USA, 2025, pp. 162-169, doi: 10.1109/ESTS62818.2025.11152450.

[5] S. Moon, **J. Park**, D. Kim, and S. Cui, "Generator Operation Strategy for Naval Vessels with MVDC Distribution in Combat Situations," *2025 IEEE Electric Ship Technologies Symposium (ESTS)*, Alexandria, VA, USA, 2025, pp. 162-169, doi: 10.1109/ESTS62818.2025.11152412.

[6] G. Park, **J. Park**, S. Cui, S.-K. Sul, "Nonlinear Optimization-Based Power-Voltage Control of Grid-Connected Converter in Weak Grid," *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Long Beach, CA, USA, 2024, pp. 228-233, doi: 10.1109/APEC48139.2024.10509166.

[7] **J. Park**, H.-J. Cho, H. Kim, S.-K. Sul, "Online Torque Control of IPMSM for Flux Weakening Operation Using Sequential Quadratic Programming," *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, Nashville, TN, USA, 2023, pp. 4750–4755, doi: 10.1109/ECCE53617.2023.10362007.

[8] **J. Park**, H.-J. Cho, J. Yun, S.-K. Sul, "Online MTPA Tracking of IPMSM Based on Min-Max Optimization," *2023 11th International Conference on Power Electronics and ECCE Asia (ICPE 2023 - ECCE Asia)*, Jeju Island, Korea, Republic of, 2023, pp. 199-204, doi: 10.23919/ICPE2023-ECCEAsia54778.2023.10213954.

## Research Project

**Development of a Medium-Voltage Drive Based on a Modular Multilevel Converter** 2023 – 2025

*HD Korea Shipbuilding and Offshore Engineering*

- Developed an EtherCAT-based control system and developed motor-control algorithms specialized for MMCs.
- Demonstrated a 3 MW operation for 4.16 kV induction motors.

**Development of a Parallel Computing Simulation Tool for Large-Scale Power Electronics Circuits** 2025 – 2026

*HD Korea Shipbuilding and Offshore Engineering*

- Developed a multicore program to execute large-scale power-electronics circuit models created in PLECS.
- Presented the work at COMPEL 2026 and transferred the technology to HD KSOE for use in MVDC shipboard system design.

**Development of a C-HIL Testbed for an LV Variable-Frequency Drive** 2026 – present

*HD Korea Shipbuilding and Offshore Engineering*

- Developed a real-time simulation model of an LVAC shipboard power system including generators, propulsion motors, thermal models for power semiconductor devices, and inverters.
- Analyzed real-time simulation model stability under discretization.
- Demonstrated an actual control system from HD KSOE without any modification.

**Development of a DC Transformer for MVDC Shipboard Power-System Integration** 2026 – present

*HD Korea Shipbuilding and Offshore Engineering*

- Developed a testing circuit to characterize the reverse-recovery behavior of high-voltage diodes. Patent application in progress.

## Awards

**Third Prize of Presentation Award**, 4th IEEE PELS Student Symposium on Power Electronics in Asia, 2026

**Excellence Prize in Creative Design Fair 2016**, Seoul National University, 2016